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Applications of Remote Sensing in Satellite Oceanography: A Review

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Abstract

Remote sensing has got wide range of applications in the field of coastal engineering, including estimation of sea surface temperature, chlorophyll content, suspended sediment concentration, yellow substance concentration, wave characteristics, potential fishing zone (PFZ) and so on. Various satellite remote sensing data has provided ream data which can be used to monitor coastal resources. The present paper gives an overview of some of the applications of remote sensing in the field of satellite oceanography. From the present review, it can be conclude that, even though IR and microwave radiometers can be used for measuring temperatures at different depths in oceans, better choice is to use microwave data as it has got the advantage of penetrating through clouds and also it gives a clear view in all weather conditions except rain. Band ratio approach can be used for identifying different water quality parameters. In recent past, satellite remote sensing has become an ideal tool for identifying Potential fishing zone (PFZ).

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1.Introduction

Now a day, remote sensing has become more and more vital in receiving data from different fields. The success of many applications of remote sensing is improved considerably by taking a multiple view approach to data collection

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i.e.data can be taken either by multistage sensing (data about a site are collected from multiple altitudes) or multispectral sensing (data are acquired simultaneously in several spectral bands) or multi temporal sensing (data are collected on more than one occasion).

It has got an important role in coastal zone management. Satellite remote sensing enable us to locate and monitor various aspects like shore line changes, bathymetry, wetland management, suspended sediment dynamics, pollution due to oil slicks, chlorophyll content etc. Earlier various conventional methods were there but it was time consuming and tedious and was difficult to see the effects due to various activities since the changes took place slowly and hence we are not able to find out the interrelated effects of various activities. By using remote sensing satellite data, we can easily study the effect of various activities in the coastal region.

Remote sensing and GIS can be effectively used in mapping and monitoring coastal resources, detecting shoreline changes, studying coastal landforms etc. Recently developed high resolution satellite data can be used to obtain the detailed land use maps at different scales for implementing various coastal regulation measures. Most of the new satellites have demonstrated their usefulness in disaster prevention, preparedness and relief.

2. Review of early works

Various studies were conducted by researchers to estimate some of the ocean characteristics and table 1 show 5 characteristics and their importance.

Surface temperature of world's ocean plays a fundamental role in the exchange of energy, momentum and moisture between ocean and atmosphere. It is a central determinant of air sea interactions and climate variability. Also, it influences the development and evolution of tropical storms and hurricanes and helps in the study of climate change and weather forecasting. Mainly infra-red (IR) and microwave radiometers are used to measure temperature at slightly different depths in the upper ocean. However IR measurements are strongly influenced by water vapour and cloud contamination, whereas microwave penetrates through cloud with little attenuation, giving an uninterrupted view of ocean surface. Rain drops having large diameter than cloud droplets will attenuate and scatter microwave radiation and hence sea surface temperature measurements (SST) retrieval is not reliable when there is rain. Microwave imagery provide nearly complete coverage where as much of the IR imagery is blocked by clouds. Also it provides insight in a number of areas including tropical instability waves, marine boundary layer dynamics and prediction of hurricane intensity (Wentz et al. 2000).

Table 1. Ocean parameters with their importance and data used.

Parameters	Data used	Importance of each parameters
Sea surface temperature	MODIS, AMSRE, TMI	Helps in the studies of climate change and weather forecasting.
Total suspended solids	DEIMOS-1, LANDSAT, ASTER	It provides information for hydrodynamic modeling of coastal zones.
Chlorophyll content	Seawifs, IKONOS, IRS P4 OCM	It indicates the presence of phytoplankton and helps in identifying PFZ.
Potential fishing zone	NOAA AVHRR, IRS P4 OCM	Helps fishermen to increase fish catch, to reduce fuel costs etc.
Wave height and spectra	GEOSAT, ASAR (ENVISAT)	Provides information for applications like offshore operational forecasting, routine navigation etc.

Chlorophyll concentration can be measured using remote sensing. Most of the studies of chlorophyll in water are based on empirical relationships between (radiance) or reflectance in narrow bands (band ratio) and chlorophyll. Various field data are collected to calibrate the statistical relationship or to validate the developed model. It was

found that the in situ measurements give a spectrum with increasing reflectance with increased chlorophyll concentration. A variety of algorithms have been used to map chlorophyll concentration in ocean. (Schalles et al. 1997). An algorithm was developed based on aircraft measurements to determine seasonal patterns of chlorophyll content.

$$\text{Log}_{10}[\text{chlorophyll}] = a + b (-\log_{10}G) \quad (1)$$

Where a and b are empirical constants derived from in situ measurements and G is $[(R_2)^2/(R_1)(R_3)]$, where R_1 is the radiance at 490nm and R_3 at 520nm and using this algorithm total chlorophyll content in the Chesapeake Bay has been mapped. (Harding et al. (1995))

Coloured dissolved organic matter (CDOM), commonly called yellow substance or Gelbstoff is the coloured fraction of the dissolved organic matter (DOM) pool in sea. Marine yellow substance originates mainly from the degradation of marine organic matter. It is found that CDOM is responsible for increasing absorption of light with decreasing wavelength in the range of 350nm- 700nm. It is considered as an important variable used for sea land interaction studies and for water environment monitoring. Many researchers have shown the capability to estimate yellow substance from satellite remote sensing data (Tassan1994, Doerffer and Fischer(1994), Arenz et al. (1996), Morel et al. (1981).

Suspended sediment (SS) increase the radiance emergent from surface waters in the visible and infrared (NIR) portion of EM spectrum. Spectral sensors on board, air craft and satellite platforms have all been used to study suspended sediment patterns. Using in situ studies it was found that wavelength between 700 and 800nm are most useful for determining SS in sea water. Many studies have developed empirical relationships between concentration of SS and radiance/ reflectance for a specified date and time. One of the study shows that an algorithm developed for one year was applicable for several years until some watershed events changes the quality (size, shape etc..) of sediments delivered to oceans (Ritchie et al. (1976), Schiebe et al. (1997).

Initially National Oceanic and Atmospheric Administration- Advanced Very High Resolution Radiometer (NOAA-AVHRR) data for sea surface temperature was utilized for finding out thermal fronts and these thermal fronts were projected as potential fishing zone (PFZ). PFZ validation program received a boost with Ocean Colour Monitoring (OCM) as satellite remote sensing techniques were found capable of obtaining information on water quality parameters like turbidity, suspended sediments, chlorophyll concentration and presence of algal blooms and large sized aquatic weeds. PFZ advisories generated from satellite retrieved SST and chlorophyll are found to be good indicators of availability as well as abundance of pelagic fishes which help fishermen to locate fishable concentration by reducing fuel costs and overall cost of operations (Solanki et al. 2003) .

A multi hazard vulnerability map (MHVM) incorporate vulnerability in understanding the risk due to a hazard. The main purpose of MVHM is to represent vulnerability, risk and hazard information together on a single map. It is referred to as a composite, synthesized and an overlaid hazard map. This helps in creating awareness in mitigating multiple hazards. The hydrological factors that influence coastal hazards are tropical cyclones, sea level rise, coastal erosion and storm surge. It was found that storm surges with tropical cyclones are most damaging parameters. About 90% of the damage is due to inundation of land by sea water. (Dube et al. (2009), Mahendra and Mohenty (2011).

Bathymetry refers to the depth of sea floor relative to sea level. It is measured using remote sensing methods that investigate sea floor indirectly without making physical contact. Most of the methods are based on the concept of using time to infer distance. Sensors emit a beam of sound, light or radiowaves and measure the round trip travel time for the beam to get reflected from a surface and return back to the surface. The time elapsed is then related to the distance travelled by the beam and is used to infer bathymetry. Shore line erosion is the temporary or permanent loss of sand from the beach or dune systems. It is caused mainly by wind driven waves. One of them used shore line monitoring using medium resolution satellite imagery and they found that this can be used to map and monitor large and dynamic shore line change along the coast of Ghana. Another study used satellite data covering low tide period.

During low tide, maximum land is exposed and even low water line and high water line are clearly visible (Ramesh et al.(2013), Chauhan and Nayak (1995).

Land use refers to the importance of land as key and finite resources for most of the human activities including agriculture, industry, recreation etc. Land cover is that part of land created by nature like forest, water bodies, river etc. One of them had made a study on demarked changes in coastal zone of Chidambaram Taluk, Tamilnadu. They performed LU/LC classification using Landsat image data and found that built up area class is the fastest growing class from 1973 to 2009 and this growth is due to population expansion and urbanisation and LU/LC map shows that area of cultivation is higher than others (Kuldeep et al.(2013)) .

Wetlands are transition zones that occupy an intermediate position between dry land and open water. The spatial distributions of Normalized difference vegetation index (NDVI) values are evaluated to determine the cut off points for waterbodies and wetted areas. Some research has been done by using lidar and radar data but majority has been concentrated on Landsat systems. One of them had tested multispectral and hyper spectral images for wetland classification using different classifiers. Mangroves are relatively easy to classify because they had a unique spectral reflectance. Hardisky et al. (1986), Kushwaha et al. (2000)). Coastal zone represents varied and highly productive systems such as mangroves, coral reefs, sea grasses and sand dunes.

A coastal regulation zone (CRZ) notification directs the administration of coastal states to prepare coastal zone management plan (CZMP) for identifying and classifying areas CRZ areas along the coast. The combination of moderate and high resolution data provided detailed coastal land use maps for implementing coastal regulatory measures (Iqbal (2012)). The available good quality ocean current data can be used to formulate the problem of routing an ocean vessel through currents. Proper ship routing minimizes fuel consumption (Hong and Mark (1994), Hagiwara (1985). Port planning should consider elements such as port development strategy, relation between port and city, economic impact and environmental impact. In one study, they have developed a fuzzy set theory to describe and evaluate the associated risk factors within the ports and terminals (Charles et al. (2012)).

3. Discussion

Results of various studies mentioned above proved that remote sensing can be effectively used to locate and monitor various aspects in ocean engineering. Reynolds et al. (2004) used microwave remote sensing for calculating SST because of its advantages over infra red SST retrievals. He made use of Tropical Rainfall Measuring Mission (TRMM) microwave imager (TMI), which is capable of measuring SST through clouds.

Deo et al. (1997) has used disaggregation models to develop an algorithm to get small interval values of wave heights from their large interval observation by breaking annual series into monthly, monthly into weekly and so on. Marcello et al. (2012) used a methodology for measuring wave velocity directly from SPOT-5 satellite image. For that he use panchromatic and multispectral datasets and the proposed method has removed the temporal lag between simultaneous panchromatic and multispectral acquisitions. Jian-Guo Li and Andrew Saulter(2012)made an indirect comparison of the updated ASAR (Advanced Synthetic Aperture Radar) wave spectra with altimeter and buoy observation for 1 year using a wave model. Along with significant wave height (SWH), he used 4 spectral sub range wave height to find the spectral characteristics of these observations and observed that this updated ASAR has got good relationship with the 2 independent observations.

Deyonget al. (2009) has developed an algorithm based on in situ measured reflectance to detect suspended matter concentration in inland water. He then applied this algorithm into a new hyper spectral data for mapping spatio-temporal distribution and the results shows that this is effective in detecting suspended matter. Also he used chl a/TSM (Total Suspended Matter) ratio to find the percentage of different constituents in total suspended matter. Isabel et al. (2013) used DEIMOS-1 satellite sensor which provide multispectral image of medium resolution for estimating TSS in Guadalquivir estuary and developed an algorithm based on relation between TSS and NIR band. They found NIR band as a good indicator of TSS. They observed that high sediment load is in the area where water from surrounding wetlands enters and lower concentrations along the coastline. Hu (2006) has made a phase

shift between temporal patterns of phytoplankton pigments and CDOM in North Atlantic Ocean using high resolution satellite data and a proposed algorithm. This phase shift indicates that CDOM in marine environment is derived from phytoplankton degradation.

Solanki et al. (2003) has developed an approach for locating PFZ using IRS P4 OCM(ocean colour monitor) derived chlorophyll concentration and AVHRR derived SST retrievals. Osawa (2010) has made a study of PFZ around Indonesia. From the obtained results it was clear that water mass moves from Java sea, which is characterised by high SST and high chlorophyll concentration during wind period and this leads to high fish catch as migration of pelagic fish takes place.

4. Conclusions

From the present literature review it can be concluded that remote sensing and GIS can be used in the mapping and monitoring of coastal resources, detecting shoreline changes, studying coastal landforms etc. Recently developed high resolution satellite data can be used to obtain the detailed land use maps (with a greater accuracy) at different scales for implementing various coastal regulation measures. Various studies are still going on to overcome the limitations of existing algorithms. Following points are drawn from the review.

- Even though IR and microwave radiometers can be used for measuring temperatures at different depths in oceans, it is better to use microwave as it has got the advantage of penetrating through clouds and also it gives a clear view in all weather conditions except rain. But in one of the paper they suggest that the use of both TMI and AVHRR can improve the accuracy of the analysis in some areas in order to take the advantage of both.
- Potential fishing zone (PFZ) can be identified using SST and chlorophyll a concentration at sea surface and indicates the availability of pelagic fish.
- Band ratio approach can be effectively used for identifying different water quality parameters.
- Satellite images taken at different dates for the same area can be used to demarcate the shoreline positions that are important for the development of ICZMP.
- All types of wetlands can be effectively studied and detailed and accurate wetland maps can be updated using satellite imagery.
- Recently developed high resolution satellite data like worldview 2 can be used to map mangroves and other coastal vegetation for the entire coastline and help us to prepare various management measures for conserving our ecosystem.
- LU/LC changes can be evaluated using remote sensing and classified images can be generated with an accuracy of almost greater than 80%.

5. References

- Arenz, R. F., Lewis, W. M., Saunders, J. F., 1996. Determination of chlorophyll and dissolved organic carbon from reflectance data for Colorado reservoirs. *International Journal of Remote Sensing* 17, 1547–1566
- Charles Roberts, Mokhtari, Kambiz, Jun Ren, Jin Wang., 2012. Decision support framework for risk management on sea ports and terminals using fuzzy set theory and evidential reasoning approach. *Expert Systems with Applications* 39, 5087–5103.
- Chauhan, P., Nayak, S., 1995. Shoreline Change – mapping from space: a case study on the Indian coast. *Inter. In the procedia of the Workshop on International Mapping from Space IRS & ISPRS WG IV/2*, 130–140.
- Chuanmin, Hu., 2006. Ocean colour reveals phase shift between marine plants and yellow substance. *IEEE Geoscience and Remote Sensing Letters* vol. 3, no. 2
- Deo, M.C., Masood Sherief, T and Abhijit Sarkar., 1997. Wave height estimation using disaggregation models. *Journal of waterway, port, coastal, and ocean engineering* 63
- Deyong Sun, Yunmei Li, Qiao Wang., 2009. A unified model for remotely estimating chlorophyll a in lake Taihu, China, Based on SVM and In Situ Hyperspectral Data. *IEEE transactions on Geoscience and Remote Sensing* vol. 47, no. 8
- Doerffer, R., Fischer, J., 1994. Concentration of chlorophyll, suspended matter, and gelbstoff in case ii waters derived from satellite coastal zone color scanner with inverse methods. *Journal of Geophysical Research* 99, 7457–7466.
- Dube, S.K., Indu Jain, Rao, A.D., Murty, T.S., 2009. Storm surge modelling for the Bay of Bengal and Arabian Sea. *Natural Hazards* 51, 3–27
- Hagiwara, H., 1985. A study on the minimum fuel consumption route. Simulation in the North Pacific Ocean. *Journal of Japan Institute of Navigation* 72, 87–96.

- Harding, L. W., Jr., Magnuson, A. Mallonee, M.E., 2005. SeaWiFS retrievals of chlorophyll in Chesapeake Bay and the Mid-Atlantic bight. *Estuarine Coastal Shelf Science* 62, 75 – 94
- Hardisky, M. A., Gross, M.F., Klemas, V., 1986. Remote sensing of coastal wetlands. *Bioscience* 36, 453-460
- Hong K. Lo., Mark McCord, R., 1998. Adaptive ship routing through stochastic ocean currents: general formulations and empirical results. *Transportation Research Part A: Policy and Practice* 32:7, 547-561.
- Isabel Caballero, Edward Morris, P., Javier Ruiz, Gabriel Navarro., 2013. Assessment of suspended solids in the Guadalquivir estuary using new DEIMOS-1 medium spatial resolution imagery. *Remote Sensing of Environment*
- Jian-Guo Li, Andrew Saulter., 2012. Assessment of the updated Envisat ASAR ocean surface wave spectra with buoy and altimeter data. *Remote Sensing of Environment* 126, 72–83
- Kuldeep Singh, Vinoth Kumar, M., Rajaprian, K., Kumar, R.S., 2013. Identification of land use land cover changes in coastal zone of Chidambaram Taluk, Tamilnadu with the help of remote sensing and GIS technology. *Elixir International Journal* 62, 17626-17629.
- Kushwaha, S. P. S., Dwivedi, R. S., Rao, B. R. M., 2000. Evaluation of various digital image processing techniques for detection of coastal wetlands using ERS-1 SAR data. *International Journal for Remote Sensing* 21, 565-579
- Mahendra, R.S., Mohanty, P.C., Bisoyi, H., Srinivasa Kumar, T., Nayak, S.R., 2011. Assessment and management of coastal multi-hazard vulnerability along the Cuddalore-Villupuram, East Coast of India using geospatial techniques. *Ocean and Coastal Management* 54, 302–311
- Marcello de Michele, Sébastien Leprince, Jérôme Thiébot, Daniel Raucoules, Renaud Binet., 2012. Direct measurement of ocean waves velocity field from a single SPOT-5 dataset. *Remote Sensing of Environment* 119, 266–271.
- Morel, A., Bricaud, A., L. Prieur., 1981. Absorption by dissolved organic matter in the sea (yellow substance) in the UV and visible domains. *Limnological Oceanography* 26, 43-53.
- Ramesh, R., Saxena, S., Purvaja, R., Suganya, G. M. D., 2013. Coastal hazard mapping in the Cuddalore region, South India. *Natural Hazards* 66, 1519-1536.
- Ritchie, J. C., Schiebe, F. R., McHenry, J. R., 1976. Remote Sensing of Suspended Sediments in Surface Water. *Photogrammetric Engineering and Remote Sensing* 42, 1539-1545.
- Reynolds, R.W., Gentemann, C.L., Wentz, F.J., 2004. Impact of TRMM SSTs on a climate-scale SST analysis. *Journal of Climate* 17, 2938–2952.
- Schalles, J.F., Schiebe, F.R., Starks, P.J., Troeger, W.W., 1997. Estimation of algal and suspended sediment loads (singly and combined) using hyperspectral sensors and integrated mesocosm experiments. *Procedia of the Fourth International Conference on Remote Sensing of Marine and Coastal Environments* 1, 247-258.
- Solanki, H. U., Dwivedi, R. M., Nayak, S. R., Somvanshi, V. S., Gulati, D. K., Pattnayak, S. K. 2003. Fishery forecast using OCM chlorophyll concentration and AVHRR SST: validation results off Gujarat coast, India. *International Journal of Remote Sensing*, 24(18), 3691-3699.
- Takashi Osawa., 2010. Study of fishery ground around Indonesia Archipelago using remote sensing data. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Science*, Volume XXXVIII, Part 8.
- Tassan. S., (1994). Local algorithms using SeaWiFS data for retrieval of phytoplankton pigments, suspended and yellow substance in coastal waters. *Applied Optics* 33, 2369-2378
- Vazeem Iqbal., 2012. Integrated Coastal Zone Management for South Andaman Coast Using Remote Sensing and Geographical Information System. *International Journal of Scientific and Research Publications* Volume 2, Issue 10.
- Wentz, F. J., Gentemann, C.L., Smith, D.K., Chelton, D.B., 2000. Satellite Measurements of Sea Surface Temperature Through Clouds. *Science*, 288(5467), 847-850.